

Utpal Kant Sharma

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Guwahati, Assam

PROFESSIONAL SUMMARY

Aspiring AI Engineer with hands-on expertise in architecting multi-agent workflows, RAG systems, and LLM orchestration. Adept at developing scalable backend architectures and integrating sophisticated machine learning models to solve complex operational challenges, with a strong focus on driving automation and actionable intelligence for B2B applications.

TECHNICAL SKILLS

Languages: Python, SQL, C.

AI & LLM Frameworks: LangChain, LangGraph, Agentic AI, RAG, Prompt Engineering, Fine-tuning, MCP.

Models & Vector DBs: OpenAI, Gemini, Claude, Groq, Hugging Face, ChromaDB, Qdrant.

Backend & Cloud: Django, DRF, FastAPI, REST APIs, PostgreSQL, Git, GitHub, Docker, AWS (EC2, S3), Vercel.

EXPERIENCE

Assam Electronics Development Corporation Ltd (Amtron)

Assam, India

Data Science Intern

Dec 2024 – Jan 2025

- Optimized complex SQL queries involving JOIN operations, aggregations, and composite indexing across 10,000+ record relational databases, reducing query execution time from 45 seconds to under 15 seconds.
- Developed ML pipelines using Python, Pandas, and NumPy to process structured datasets and extract operational insights.
- Built Matplotlib dashboards to visualize quarterly operational KPIs and translate SQL outputs into internal reports.

PROJECTS

Smart City Feedback Management System (SEVA) *Django, PostgreSQL, PostGIS, Gemini API, FastAPI*

- Architected a full-stack civic engagement platform integrating the Gemini API and Hugging Face Transformers for automated complaint classification.
- Benchmarked system performance, achieving 87% severity and 92% category prediction accuracy on test datasets to automate triage routing.
- Implemented PostGIS geo-spatial indexing to cluster location-based data, generating interactive heat maps for high-priority civic zones.

NeuroLeads AI: Multi-Agent B2B Sales Orchestrator *LangGraph, LangChain, PostgreSQL, Docker*

- Built a multi-agent LLM pipeline using LangGraph with conditional routing for research, structured extraction, drafting, and a self-critique loop.
- Engineered defensive parsing to handle malformed non-JSON LLM responses and configured reasoning-effort parameters to optimize API token consumption.
- Containerized a FastAPI backend with Postgres-backed LangGraph checkpointing to enable a persistent, human-in-the-loop approval workflow across separate API requests.

Multi-Agent Research Assistant

LangChain, Tavily API, Streamlit

- Orchestrated a 4-stage search, reader, writer, and critic pipeline gathering web sources via the Tavily API.
- Built a 3-tier fallback scraping strategy (trafilatura, readability, BeautifulSoup) and exposed the pipeline through a custom Streamlit UI for live status tracking.

EDUCATION

The Assam Kaziranga University

Guwahati, Assam

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.08/10

Aug 2022 – May 2026

- Coursework:** Data Structures & Algorithms, Machine Learning, Database Management Systems, Operating Systems, Computer Networks.

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate (Coursera Credential ID: YB15NDMCH302) – Covered RAG architectures, agentic workflows, CrewAI, AutoGen, and MCP.